

- BC 1. What are all values of x for which the function f defined by $f(x) = x^3 + 3x^2 - 9x + 7$ is increasing?

1998

NC

- (A) $-3 < x < 1$
(B) $-1 < x < 1$
(C) $x < -3$ or $x > 1$
(D) $x < -1$ or $x > 3$
(E) All real numbers

- BC 27. If $f(x) = \frac{1}{3}x^3 - 4x^2 + 12x - 5$ and the domain is the set of all x such that $0 \leq x \leq 9$, then the absolute maximum value of the function f occurs when x is

1973

NC

- (A) 0 (B) 2 (C) 4 (D) 6 (E) 9

- BC 3. The function f given by $f(x) = 3x^5 - 4x^3 - 3x$ has a relative maximum at $x =$

1997

NC

- (A) -1 (B) $-\frac{\sqrt{5}}{5}$ (C) 0 (D) $\frac{\sqrt{5}}{5}$ (E) 1

- BC 14. The derivative of f is $x^4(x-2)(x+3)$. At how many points will the graph of f have a relative maximum?

1993

SC

- (A) None (B) One (C) Two (D) Three (E) Four

- BC 2. At what values of x does $f(x) = 3x^5 - 5x^3 + 15$ have a relative maximum?

1985

NC

- (A) -1 only (B) 0 only (C) 1 only (D) -1 and 1 only (E) -1, 0 and 1

- BC 19. A polynomial $p(x)$ has a relative maximum at $(-2, 4)$, a relative minimum at $(1, 1)$, a relative maximum at $(5, 7)$ and no other critical points. How many zeros does $p(x)$ have?

1988

NC

- (A) One (B) Two (C) Three (D) Four (E) Five

- BC 2. If $f(x) = 2x^2 + 1$, then $\lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x^2}$ is

1993

SC

- (A) 0 (B) 1 (C) 2 (D) 4 (E) nonexistent